

```

# =====
#          🛡️ CRYPTOVAULT - URL SECURITY ANALYZER
#          TASK 5 - PROSTACKHUB
# =====

from flask import Flask, request, render_template_string
from urllib.parse import urlparse
import ipaddress
import re

app = Flask(__name__)

# =====
#          HTML + CSS DESIGN
# =====

HTML = """
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>CryptoVault | URL Security Analyzer</title>

  <style>

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
      font-family: Arial, Helvetica, sans-serif;
    }

    body {
      min-height: 100vh;
      background:
        radial-gradient(circle at top left, #172554, transparent 35%),
        radial-gradient(circle at bottom right, #312e81, transparent 35%),
        #050816;
      color: white;
      padding: 30px 15px;
    }

    .container {
      max-width: 950px;
      margin: auto;
    }
  </style>

```

```
.header {
  text-align: center;
  padding: 25px;
  margin-bottom: 25px;
}

.logo {
  font-size: 55px;
  margin-bottom: 10px;
}

.title {
  font-size: 38px;
  font-weight: bold;
  background: linear-gradient(90deg, #60a5fa, #c084fc);
  -webkit-background-clip: text;
  color: transparent;
}

.subtitle {
  color: #b8c1d9;
  margin-top: 10px;
  font-size: 16px;
}

.card {
  background: rgba(15, 23, 42, 0.88);
  border: 1px solid rgba(148, 163, 184, 0.25);
  border-radius: 22px;
  padding: 30px;
  box-shadow: 0 20px 50px rgba(0,0,0,0.4);
  margin-bottom: 25px;
}

label {
  display: block;
  margin-bottom: 10px;
  font-weight: bold;
  color: #e2e8f0;
}

.url-box {
  display: flex;
  gap: 10px;
}

input {
  flex: 1;
  padding: 16px;
  border-radius: 12px;
```

```

    border: 1px solid #475569;
    background: #0f172a;
    color: white;
    outline: none;
    font-size: 15px;
}

input:focus {
    border-color: #60a5fa;
    box-shadow: 0 0 12px rgba(96,165,250,.25);
}

button {
    padding: 15px 25px;
    border: none;
    border-radius: 12px;
    background: linear-gradient(135deg, #2563eb, #7c3aed);
    color: white;
    font-weight: bold;
    cursor: pointer;
    transition: .3s;
}

button:hover {
    transform: translateY(-2px);
    box-shadow: 0 10px 25px rgba(99,102,241,.35);
}

.score-card {
    text-align: center;
    padding: 25px;
    border-radius: 18px;
    background: rgba(30,41,59,.8);
    margin-bottom: 20px;
}

.score {
    font-size: 65px;
    font-weight: bold;
}

.safe {
    color: #22c55e;
}

.medium {
    color: #facc15;
}

.danger {

```

```
        color: #ef4444;
    }

    .status {
        font-size: 20px;
        font-weight: bold;
        margin-top: 5px;
    }

    .grid {
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 15px;
        margin-top: 20px;
    }

    .result {
        padding: 18px;
        border-radius: 14px;
        background: #111827;
        border: 1px solid #334155;
    }

    .result-title {
        color: #94a3b8;
        font-size: 13px;
        margin-bottom: 7px;
    }

    .result-value {
        font-weight: bold;
        font-size: 17px;
    }

    .recommendations {
        margin-top: 20px;
        padding: 20px;
        border-radius: 15px;
        background: rgba(30,41,59,.75);
    }

    .recommendations h3 {
        color: #60a5fa;
        margin-bottom: 12px;
    }

    .recommendations li {
        margin: 9px 0;
        color: #dbeafe;
        list-style: none;
    }
```

```

}

.recommendations li::before {
  content: "⚠ ";
  color: #facc15;
}

.footer {
  text-align: center;
  color: #64748b;
  padding: 20px;
  font-size: 13px;
}

.error {
  background: rgba(127,29,29,.3);
  color: #fca5a5;
  padding: 15px;
  border-radius: 12px;
  margin-top: 15px;
}

@media(max-width: 650px) {

  .title {
    font-size: 28px;
  }

  .url-box {
    flex-direction: column;
  }

  .grid {
    grid-template-columns: 1fr;
  }

  button {
    width: 100%;
  }
}

```

</style>

</head>

<body>

<div class="container">

<div class="header">

```

<div class="logo"><img alt="CryptoVault logo" data-bbox="344 108 358 122"/></div>

<div class="title">
  CRYPTOVAULT
</div>

<div class="subtitle">
  Smart URL Security & Risk Analyzer
</div>

</div>

<div class="card">

  <form method="POST">

    <label>
      <img alt="key icon" data-bbox="245 400 260 415"/> Enter a URL to analyze
    </label>

    <div class="url-box">

      <input
        type="text"
        name="url"
        placeholder="https://example.com"
        value="{{ url }}"
        required
      >

      <button type="submit">
        <img alt="magnifying glass icon" data-bbox="285 625 300 640"/> Analyze URL
      </button>

    </div>

  </form>

  {% if error %}

    <div class="error">
      <img alt="error icon" data-bbox="245 785 260 800"/> {{ error }}
    </div>

  {% endif %}

</div>

```

```
{% if result %}

<div class="card">

    <div class="score-card">

        <div class="score {{ result.color }}">
            {{ result.score }}/100
        </div>

        <div class="status">
            {{ result.status }}
        </div>

    </div>
```

```
<h2>🔒 Security Analysis</h2>
```

```
<div class="grid">

    <div class="result">

        <div class="result-title">
            HTTPS Protection
        </div>

        <div class="result-value">
            {{ result.https }}
        </div>

    </div>
```

```
<div class="result">

    <div class="result-title">
        IP Address Detected
    </div>

    <div class="result-value">
        {{ result.ip }}
    </div>

</div>
```

```
<div class="result">
```

```
<div class="result-title">
    Suspicious Characters
</div>

<div class="result-value">
    {{ result.suspicious }}
</div>

</div>

<div class="result">

    <div class="result-title">
        URL Length
    </div>

    <div class="result-value">
        {{ result.length }} characters
    </div>

</div>

<div class="result">

    <div class="result-title">
        Domain
    </div>

    <div class="result-value">
        {{ result.domain }}
    </div>

</div>

<div class="result">

    <div class="result-title">
        URL Scheme
    </div>

    <div class="result-value">
        {{ result.scheme }}
    </div>

</div>
```

```
</div>
```

```
<div class="recommendations">
```

```
<h3>🔒 Security Recommendations</h3>
```

```
<ul>
```

```
{% for recommendation in result.recommendations %}
```

```
<li>{{ recommendation }}</li>
```

```
{% endfor %}
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
{% endif %}
```

```
<div class="footer">
```

```
🔒 CryptoVault URL Security Analyzer
```

```
<br><br>
```

```
MADE by Aiza Eman – Software Engineer
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

```
"""
```

```
# =====
```

```
# URL ANALYSIS FUNCTION
```

```
# =====
```

```
def analyze_url(url):
```

```
    result = {}
```

```
    # -----
```

```
    # Clean URL
```

```

# -----

url = url.strip()

if not url.startswith(("http://", "https://")):
    url = "https://" + url

parsed = urlparse(url)

# -----
# Domain
# -----

domain = parsed.hostname

if not domain:
    raise ValueError("Invalid URL. Please enter a valid website address.")

result["domain"] = domain
result["scheme"] = parsed.scheme.upper()

score = 100
recommendations = []

# -----
# HTTPS CHECK
# -----

if parsed.scheme.lower() == "https":

    result["https"] = "✅ Enabled"

else:

    result["https"] = "❌ Not Secure"
    score -= 30

    recommendations.append(
        "Use HTTPS websites whenever possible."
    )

# -----
# IP ADDRESS CHECK
# -----

ip_detected = False

try:

    ipaddress.ip_address(domain)

```

```

        ip_detected = True
except ValueError:
    ip_detected = False

if ip_detected:
    result["ip"] = "⚠ Yes"
    score -= 20

    recommendations.append(
        "The URL uses an IP address instead of a normal domain."
    )
else:
    result["ip"] = "✅ No"

# -----
# SUSPICIOUS CHARACTER CHECK
# -----

suspicious_pattern = r"[@<>\"'`|{}\\]"

suspicious_found = re.findall(
    suspicious_pattern,
    url
)

if suspicious_found:
    result["suspicious"] = (
        f"⚠ Yes ({len(suspicious_found)} found)"
    )

    score -= min(
        20,
        len(suspicious_found) * 5
    )

    recommendations.append(
        "Review unusual characters carefully before visiting the URL."
    )
else:
    result["suspicious"] = "✅ No"

# -----

```

```

# URL LENGTH CHECK
# -----

url_length = len(url)

result["length"] = url_length

if url_length > 200:

    score -= 15

    recommendations.append(
        "The URL is unusually long. Verify that all parameters are expected."
    )

elif url_length > 100:

    score -= 5

    recommendations.append(
        "The URL is moderately long; review its parameters carefully."
    )

# -----
# SUSPICIOUS KEYWORDS
# -----

suspicious_words = [
    "login",
    "verify",
    "verification",
    "password",
    "account",
    "update",
    "secure",
    "confirm"
]

found_words = []

for word in suspicious_words:

    if word in url.lower():

        found_words.append(word)

if len(found_words) >= 3:

    score -= 10

```

```

        recommendations.append(
            "The URL contains several security-related keywords. "
            "Verify the website through an official source."
        )

# -----
# QUERY PARAMETER CHECK
# -----

if parsed.query:

    if len(parsed.query) > 150:

        score -= 5

        recommendations.append(
            "The URL contains a large query string. "
            "Check its parameters before proceeding."
        )

# -----
# FINAL SCORE
# -----

score = max(0, min(100, score))

if score >= 80:

    status = "🟢 LOW RISK"
    color = "safe"

    if not recommendations:

        recommendations.append(
            "No obvious warning signs were detected by this basic analyzer."
        )

elif score >= 50:

    status = "🟡 MEDIUM RISK"
    color = "medium"

else:

    status = "🔴 HIGH RISK"
    color = "danger"

result["score"] = score
result["status"] = status
result["color"] = color

```

```
result["recommendations"] = recommendations
```

```
return result
```

```
# =====  
#                               FLASK ROUTE  
# =====
```

```
@app.route("/", methods=["GET", "POST"])  
def home():
```

```
    result = None  
    error = ""  
    url = ""
```

```
    if request.method == "POST":
```

```
        url = request.form.get("url", "").strip()
```

```
        if not url:
```

```
            error = "Please enter a URL."
```

```
        else:
```

```
            try:
```

```
                result = analyze_url(url)
```

```
            except ValueError as e:
```

```
                error = str(e)
```

```
            except Exception:
```

```
                error = "Unable to analyze this URL."
```

```
    return render_template_string(  
        HTML,
```

```
        result=result,
```

```
        error=error,
```

```
        url=url
```

```
)
```

```
# =====  
#                               APPLICATION  
# =====
```

```
if __name__ == "__main__":  
  
    print("=" * 60)  
    print("🔒 CRYPTOVAULT - URL SECURITY ANALYZER")  
    print("=" * 60)  
    print("🌐 Open your browser and visit:")  
    print("👉 http://127.0.0.1:5002")  
    print("=" * 60)  
  
    app.run(  
        host="127.0.0.1",  
        port=5002,  
        debug=True  
    )
```